

7.8 Quiz: Logarithms (no calculator)

1. [Maximum mark: 5]

Write each of the following expressions in the form $\ln k$, where $k \in \mathbb{Z}^+$.

(a) $\ln 5 + \ln 6$ [1]

(b) $2 \ln 3$ [2]

(c) $-\ln \frac{1}{4}$ [2]

2. [Maximum mark: 5]

Let $\log_{10} 2 = p$ and $\log_{10} 5 = q$.

(a) Find an expression for $\log_{10} 40$ in terms of p and q . [3]

(b) Find an expression for $\log_5 4$ in terms of p and q . [2]

3. [Maximum mark: 5]

It is given that $\log_{10} a = \frac{1}{2}$, where $a > 0$.

Find the value of

$$(a) \log_{10}\left(\frac{1}{a}\right); \quad [2]$$

(b) $\log_{100} a$. [3]

[illegible]

4. [Maximum mark: 5]

Solve the equation $2 \ln x = \ln 25 + 6$. Give your answer in the form $x = pe^q$ where $p, q \in \mathbb{Z}^+$.